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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination - APRIL 2021

AMT202 – ALLIED MATHEMATICS – II

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Obtain the equation, one of its root is $\sqrt{3}+\sqrt{5}$.

2. Find the Real part of $\cos(x+iy)$.

3. $\int_0^1 \int_0^2 \int_0^3 xyz\,dx\,dy\,dz$= ?

4. Form the partial differential equation by eliminating the arbitrary constants from $z = (x^2+a)(y^2+b)$.

5. Find the divergence of $\vec{F} = x^2y\vec{i} + xz\vec{j} + 2yz\vec{k}$.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Solve $x^4-6x^3+11x^2-10x+2=0$ given that $2-\sqrt{3}$ is a root.

7. Show that $\cosh^{-1} x = \log(x + \sqrt{x^2-1})$.

8. Evaluate $\iint xy\,dxdy$ over the region bounded by x-axis, ordinate x = 2a and the curve $x^2 = 4ay$.

9. Solve : p + q = sin x + sin y.

10. Find the directional derivative of the function yz+zx+xy in the direction $2\vec{i}-3\vec{k}$ at(1, 2, 3).

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) Solve the equation $x^3-12x^2+39x-28=0$ are in A.P.

(or)

b) Solve: $6x^6 - 35x^5 + 56x^4 - 56x^2 + 35x - 6 = 0$.

12. a) Express cos 8θ as a power of cos θ.

(or)

b) If cos(x+iy) = cosθ + i sin θ, then show that cos 2x + cosh 2y = 2.

13. a) Evaluate: $\int_0^{4a} \int_{\frac{x^2}{4a}}^{x} x^2 y \, dx \, dy$.

(or)

b) Evaluate $\int_0^a \int_x^a \left(x^2 + y^2\right) dxdy$.

14. a) Obtain the complete and singular solutions of $z = px + qy + p^2 + pq + q^2$.

(or)

b) Solve : $x(y-z)p + y(z-x)q = z(x-y)$.

15. a) Verify Stoke's theorem $\int \vec{F}_C \cdot d\vec{r}$ where $\vec{F} = (2x-y)\vec{i} - yz^2\vec{j} - y^2z\vec{k}$ where S is the upper half of the sphere $x^2 + y^2 + z^2 = 1$ and 'C' is the boundary.

(or)

b) Verify Green's theorem evaluate $\int (x^2 - y^2)dx + 2xydy$ where C is the boundary of the region bounded by the lines x = 0, x = a, y = 0, y = b.

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